

GR6211: Microeconomic Analysis I Part I

Fall 2026

Columbia University

Contact Details

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Classes: Tuesdays and Thursdays 10:10-11:25, 414 Pupin Laboratories

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Course Description

G6211 is the first semester course in the microeconomics sequence for the PhD program. It consists of two halves. I will teach the first half, which will cover choice theory, while the second half will be taught by Professor Bernard Salanie, who will cover efficiency (including efficient risk sharing), general equilibrium, market failures (public goods, externalities) and social choice.

The choice theory portion of the class will focus on the fundamental models of behavior which underlie most economic analysis. The majority of the course will focus on consumer behavior. We will begin by studying the relationship between utility maximization, preference maximization and choice, before moving on to study the properties of demand functions, which relate parameters of the consumer's problem to their choices. We will then digress briefly to discuss producer theory and provide analogous results in this domain. We will conclude by studying the standard models of choice under risk and uncertainty, and over time

Prerequisites

This graduate economics class is intended for the first-year students in PhD programs in Economics. Students from other programs must obtain permission from me and the DGS of the economics department in order to enroll in the class.

Assessment

Assessment for the first part of the course will consist of two elements: homeworks (20%) and a final (80%)

Homeworks

There will be problem sets (almost) every week. You are encouraged to work with your classmates in solving them, but I request each of you to write up and turn in your own answers. Of course if you don't work hard on problem sets in your own right you are likely to find it very difficult to do well on your exams. Solutions to the problems will be posted on the course website. Each homework will be graded from 0 – 10. Late homeworks will not be accepted or graded. However, your lowest mark across the homeworks will be discounted, so you get one free pass.

Final

In addition to homeworks, there will be a final exam. This will take place on or after my final lecture on October 22nd. Hopefully we will be able to find a room so that the exam can be 2 hours long.

Course Materials

By and large, the course will be based on slides/lecture notes, which I will make available online. I try to make use of the board as much as possible, particularly for proofs. I truly think it is hard to understand a proof if you don't write it out yourself. I will also release notes at some point after class.

The course will most closely resemble *Rubinstein, A. (2005). Lecture Notes in Microeconomic Theory: The Economic Agent*¹. This is available for free download at <http://arielrubinstein.tau.ac.il/books.html>. I strongly recommend that you download it, not least because I will be setting some homework exercises from there.

While it is not required for my part of the course, I would strongly recommend you get hold of *Mas-Colell, A., Whinston, M. D., & Green, J. R. (2006). Microeconomic theory*. This will be useful for all 4 parts of the microeconomic sequence and for the rest of your economics career.

For people who are particularly interested in the material for this section of the course I would also recommend *Kreps (1988), Notes on the Theory of Choice* (which is a genuinely beautiful and mercifully short book) and *Ok, E. (2004) Real Analysis with Economic Applications* (also beautiful, but a lot longer)

¹ This book is continually updated. The latest version is from 2025.

Course Policies

Regrading

Corrected assignments and exams will be available at most seven days after the deadline or the date of the test. Students then have at most seven days to introduce a complaint regarding a grade (that means exactly two weeks after the deadline or the date of the test or assignment submission). I deal personally with all complaints. Requests must be placed in my mailbox on the 11th floor of IAB. I expect them to be specific and motivated. This means that you must include a copy of the question(s) that need(s) to be re-graded, a copy of your answer(s) and the relevant section of the answer key, and an explanation as to why the question(s) need(s) to be re-graded. If I find the request motivated, then I will re-grade the question(s) myself (which may possibly result in a lower grade if I find it justified). My decision is final. Please do not use either my or the TAs' office hours to complain about grades.

Academic Dishonesty

As members of an academic community, each of us has a responsibility to participate in scholarly discourse and research in a manner characterized by intellectual honesty and scholarly integrity, and cheating on exams is a very serious violation. Any suspected case of cheating will be reported to the university, and students who breach their intellectual responsibility in this regard should anticipate being asked to leave Columbia.

Class Conduct

Cell phone use is not allowed. Laptops are fine for taking notes (not that I would recommend it), but please respect your classmates and instructor by limiting yourself to class-related activities. Though you may be a phenomenal multi-tasker, using a laptop for purposes other than taking notes is distracting to those around you.

Artificial Intelligence

It is highly likely that you could use ChatGPT or equivalent to help you solve homework problems. While I am not going to police this, I strongly advise you not to do it, for two reasons. The better reason is that throwing homework problems into Chat will not help you become a better economist, but actually figuring out how to solve them will. And at the end of the day that is what you are here for. By analogy, if you are training for a marathon, you could decide to use a car to do the longer training runs. No one will catch you doing it, but it's also not going to help you to complete the marathon. What does help is feeling the burn during a long training run, or equivalently, staring down a homework problem till you figure out how to crack it.

The less good (but still important) reason not to do it is that it won't help you to do well in the course. Homework counts for 20% of your grade for my part of the course, and there is generally very little variation in scores. What really matters is how you do in the exam, and seeing as this is in class, Chat will not be able to come to your aid.

Topics

1. Choice, Preference and Utility (Approx 4 lectures)
 - a. Introduction to representation theorems
 - b. Choices as preference maximization
 - c. Preference maximization as utility maximization
 - d. Extensions: infinite spaces and incomplete data
 - e. Choice from budget sets
 - f. Empirical Evidence
 - g. Stochastic choice
2. Demand Functions (Approx 2 lectures)
 - a. Utility maximization
 - b. Cost minimization
 - c. Duality
3. Producer Theory (Approx 2 lectures)
 - a. Single input case
 - b. Multi input case
4. Decision Making Under Risk (Approx 3 lectures)
 - a. Expected utility theory
 - b. The Independence axiom
 - c. Risk vs Uncertainty
 - d. Subjective Expected Utility
 - e. Updating
 - f. Empirical Evidence
5. Decision Making Over Time (Approx 2 lectures)
 - a. Exponential discounting
 - b. Empirical evidence